

10.12 Lagrange Error Bound

Calculus

Solutions

Practice

1. The third Maclaurin polynomial for $\sin x$ is given by $p(x) = x - \frac{x^3}{3!}$. If this polynomial is used to approximate $\sin(0.1)$, what is the Lagrange error bound?

Centered at $x=0$

Interval: $[0, 0.1]$

$$\frac{\max [f^{(4)}(z)] \cdot (0.1 - 0)^4}{4!}$$

$f'''' = \sin x$

$\max \text{ of } \sin x = 1$

$$\frac{1 \cdot (0.1)^4}{4!} = 4.1667 \times 10^{-6}$$

2. If the Taylor Polynomial for approximating $\cos x$ is given by $1 - \frac{x^2}{2!} + \frac{x^4}{4!}$, what is the upper bound for the error in the approximation of $\cos(0.3)$?

$$\frac{\max [f^5(z)] (0.3 - 0)^5}{5!}$$

Interval: $[0, 0.3]$

$n = 4$
centered at $x = 0$

$$f^5 = -\sin x$$

$$\max \text{ of } -\sin x = 1$$

$$\frac{1 (0.3)^5}{5!}$$

$$= 2.025 \times 10^{-5}$$

3. If the Taylor Polynomial about $x = 0$ for the approximation of e^x is given by $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!}$, what is the upper bound for the error in the approximation of e ?

$$f^6 = e^x$$

max on interval $[0, 1]$ is e

$$\frac{\max [f^6(z)] (1 - 0)^6}{6!}$$

$$\frac{e \cdot (1)^6}{6!}$$

$$= 0.00377$$

e^1 means interval $[0, 1]$
 $n = 5$

4. Let f be a function that has derivatives of all orders for all real numbers and let $P_3(x)$ be the third-degree Taylor Polynomial for f about $x = 0$. $|f^{(n)}(x)| \leq \frac{n}{n+1}$, for $1 \leq n \leq 5$ and all values of x . Of the following, which is the smallest value of k for which the Lagrange error bound guarantees that $|f(1) - P_3(1)| \leq k$?

$$R_3(x) \leq k$$

$$\frac{f^{(4)}(x) (1-0)^4}{4!} \leq k$$

$$\frac{\frac{4}{5} \cdot 1}{4!}$$

(A) $\frac{5}{6}$

(B) $\frac{5}{6} * \frac{1}{5!}$

(C) $\frac{5}{6} * \frac{1}{4!}$

(D) $\frac{4}{5} * \frac{1}{4!}$

5. The function f has derivatives of all orders for all real numbers, $f^{(4)}(x) = e^{\cos x}$. If the third-degree Taylor Polynomial for f about $x = 0$ is used to approximate f on the interval $[0, 1]$, what is the Lagrange error bound?

$$\frac{\max [f^4(z)] (1 - 0)^4}{4!}$$

$$\frac{e^1 \cdot 1}{4!}$$

$$\approx 0.11326$$

6. The Taylor series for a function f about $x = 3$ is given by $\sum_{n=0}^{\infty} (-1)^n \frac{3n+1}{2^n} (x-3)^n$ and converges to f for $0 \leq x \leq 5$. If the third-degree Taylor Polynomial for f about $x = 3$ is used to approximate $f\left(\frac{13}{4}\right)$, what is the alternating series error bound?

$$a_n = \frac{3n+1}{2^n} (x-3)^n \rightarrow a_4 = \frac{13}{16} \left(\frac{13}{4} - 3\right)^4$$

$$|R_3| \leq \frac{13}{16} \left(\frac{1}{4}\right)^4$$

7. Let f be a polynomial function with nonzero coefficients such that $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$. $T_4(x)$ is the fourth-degree Taylor Polynomial for f about $x = c$ such that $T_4 = b_0 + b_1(x-c) + b_2(x-c)^2 + b_3(x-c)^3 + b_4(x-c)^4$. Based on the Lagrange error bound, $f(x) - T_4(x)$ must equal which of the following?

$$|R_4(x)| = \frac{f^{(5)}(z)}{5!} (x-c)^5$$

$$f^{(5)}(x) = 5! a_5 \rightarrow R_4(x) = a_5 (x-c)^5$$

(A) x

(B) $(x-c)^5$

(C) $a_5(x-c)^5$

(D) $\frac{a_5(x-c)^5}{5!}$

8. Let $P(x)$ be the sixth-degree Taylor Polynomial for a function f about $x = 0$. Information about the maximum of the absolute value of selected derivatives of f over the interval $0 \leq x \leq 1.5$ is given below.

$$\max_{0 \leq x \leq 1.5} |f^{(5)}(x)| = 9.3$$

$$\max_{0 \leq x \leq 1.5} |f^{(6)}(x)| = 62.1$$

$$\max_{0 \leq x \leq 1.5} |f^{(7)}(x)| = 481.3$$

What is the smallest value of k for which the Lagrange error bound guarantees that $|f(1.5) - P(1.5)| \leq k$?

$$\frac{\max [f^{(7)}(z)] (1.5-0)^7}{7!} = \frac{(481.3)(1.5)^7}{7!} \approx 1.6316$$

9. The function f has derivatives of all orders for all real numbers. Values of f and its first four derivatives at $x = 2$ are given in the table.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(4)}(x)$
2	6	-12	18	-24	34

- a. Write the third-degree Taylor Polynomial for f about $x = 2$, and use it to approximate $f(1.5)$.

$$P_3(x) = 6 + -12(x-2) + \frac{18(x-2)^2}{2} + \frac{-24(x-2)^3}{3!}$$

$$P_3(x) = 6 - 12(x-2) + 9(x-2)^2 - 4(x-2)^3$$

$$f(1.5) \approx P_3(1.5) = 14.75$$

- b. The fourth derivative of f satisfies the inequality $|f^{(4)}(x)| \leq 48$ for all $x > 1$. Use the Lagrange error bound to show that the approximation found in part (a) differs from $f(1.5)$ by no more than $\frac{1}{8}$.

$$\begin{aligned} R_3(1.5) &\leq \frac{\max[f^{(4)}(z)](1.5-2)^4}{4!} \\ &\leq \frac{48(-0.5)^4}{4!} \\ &\leq 0.125 = \frac{1}{8} \end{aligned}$$

10. Let h be a function having derivatives of all orders for $x > 0$. Selected values for the first four derivatives of h are given for $x = 3$. Use the Lagrange error bound to show that the third-degree Taylor polynomial for h about $x = 3$ approximates $h(2.9)$ with an error less than 3×10^{-4} .

x	$h(x)$	$h'(x)$	$h''(x)$	$h'''(x)$	$h^{(4)}(x)$
3	317	$\frac{753}{4}$	$\frac{1383}{4}$	$\frac{3483}{8}$	$\frac{1125}{16}$

$$\begin{aligned} R_3(x) &\leq \frac{\max[h^{(4)}(z)](x-3)^4}{4!} \\ &\leq \left(\frac{1125}{16}\right)\left(\frac{1}{24}\right)(-0.1)^4 \\ &\leq 2.9297 \times 10^{-4} \quad \checkmark \end{aligned}$$

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Test Prep

11. Calculator allowed.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(4)}(x)$
3	4	-8	14	-22	30

The function f has derivatives of all orders for all real numbers. Values of f and its first four derivatives at $x = 3$ are given in the table.

- a. Write an equation for the line tangent to the graph of f at $x = 3$ and use it to approximate $f(2.5)$.

$$\begin{aligned} y - 4 &= -8(x - 3) \\ y &= -8x + 28 \\ f(2.5) &\approx -8(2.5) + 28 = 8 \end{aligned}$$

- b. Write the third-degree Taylor polynomial for f about $x = 3$, and use it to approximate $f(2.5)$.

$$P_3(x) = 4 + -8(x-3) + \frac{14(x-3)^2}{2} + \frac{-22(x-3)^3}{3!}$$

$$P_3(x) = 4 - 8(x-3) + 7(x-3)^2 - \frac{11}{3}(x-3)^3$$

$$f(2.5) \approx P_3(2.5) = 10.208$$

- c. Is there enough information to determine whether f has a critical point at $x = 2.5$? If not, explain why not. If so, determine whether $f(2.5)$ is a relative maximum, relative minimum, or neither, and give a reason for your answer.

There is not enough information. We don't know if $f'(2.5) = 0$ or if $f'(2.5)$ does not exist. The Taylor Polynomial only gives us an approximation of $f(x)$.

- d. The fourth derivative of f satisfies the inequality $|f^{(4)}(x)| \leq 48$ for all $x > 2$. Use the Lagrange error bound to show that the approximation found in part (b) differs from $f(2.5)$ by no more than $\frac{1}{8}$.

$$R_3(2.5) \leq \frac{\max[f^{(4)}(z)](2.5-3)^4}{4!}$$

$$\leq \frac{48(-0.5)^4}{4!}$$

$$\leq 0.125$$

- e. What is the coefficient of the $(x-3)^3$ term in the Taylor series for f' , the derivative of f , about $x = 3$?

In the Taylor Polynomial for f the coefficient of $(x-3)^4$ is $\frac{f^{(4)}(3)}{4!} = \frac{30}{24} = \frac{5}{4}$,
therefore in the Taylor Polynomial for f' , the coefficient of $(x-3)^3$ is $\frac{5}{4} * 4 = 5$